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RESEARCH ARTICLE

Intrusion Detection Model Using Fused Quantum NN–LeNet With Improved Elliptic Curve Cryptography Based Data Security

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ABSTRACT An intrusion detection system (IDS) is a network security tool originally designed to identify vulnerability exploits against a target program or computer. Once the attacker has access to the network, they can rapidly collect information. Hence, there is a need of intrusion detection system, which also extends to secure the data as well. This work focusses on a new DL-based IDS model along with data security assurance by deploying enhanced ECC model. Initially, the input data is pre-processed using min-max normalization process. Subsequently, Quantum based Hybrid model is introduced that combines the models like flatten layer proposed feature map with EVD-based QNN (FLPF with EVD-QNN) and LeNet models. Here, the final fused detection results are determined by Double Sigmoid Median min-max normalization-based Score Level Fusion (DSMMN-SLF). Subsequently to the intrusion detection, this work concentrates on secured data transmission, for which Enhanced Elliptic Curve Cryptography (EECC) is proposed that encrypts the data to assure its privacy. From analysis, the precision of detecting the intrusions using FLPF with EVD-QNN+LNt was 97% which is high when comparing to existing methods.

INDEX TERMS Intrusion detection, quantum neural networks, LeNet, elliptic curve cryptography, data security.

I. INTRODUCTION

Data security has emerged as a critical issue due to recent developments in communication and internet technologies [1], [2]. It uses technologies like firewalls, antivirus software, and IDS to ensure the network security and all of its associated assets are protected in cyberspace. One of them is a NIDS, an attack detection method that provides the essential defense by continuously examining network traffic for hostile and suspicious behavior [3], [4]. Since then, various IDS software has been developed and improved to meet network security standards. However, the speed at which technology developed has led to a huge growth in the dimension of networks and the number of applications they can support [5], [6]. As a result, several network nodes create and distribute a substantial amount of crucial data.

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The confidentiality of this kind of data and networking nodes has become a difficult problem due to the proliferation of many risks [7], [8]. Every network node is susceptible to security issues. It has been demonstrated that the present IDSs are ineffective at lowering FAR and detecting a range of threats, including zero-day attacks. To give the network security, it needs a reliable, accurate, and reasonably valued NIDS is unavoidably required [9], [10]. Thus, in the modern era, IDS are essential for ensuring the network security. Network traffic analysis and IDS monitoring are used to classify attacks into several groups [11], [12]. Numerous factors that have been collated into a database to detect different attack kinds make up the internet traffic activity. The constant increase in the massive amount of online created data is posing serious challenges for technology [13], [14]. Moreover, understanding the types of data that poses a variety of attacks is therefore essential. Recently, Cryptographic methods are used to secure data by encrypting it and

rendering it illegible [15], [16]. This cryptography allows data to be decrypted and encrypted without the help of the sender. It also provides user authentication. Only with the aid of enhanced communication networks, the encrypted data can be transmitted across the Internet. The best way to provide the required protection against hackers is through encryption, which also helps to prevent unsafe cloud communication. The asymmetric key cryptography technique employs two distinct keys, private and public keys for the decryption and encryption procedures. In this setup, every user has their own public key and private key. If the user (sender) provides the actual data as encoded files (secret code text) to any specific user (receiver), the receiver gets the data and uses their own public key to decode it back to the original content. The key exchange process is simple, but the procedures for decryption and encryption are difficult and time-consuming. A few examples of practical public key protocols for key sharing include Diffie-Hellman, ECC, and RSA [17]. Thereby, a new IDS model is proposed with EECC-based data security that offers below contributions. Proposes flatten layer proposed feature map with EVD-based QNN for detecting the intrusion in network. The proposed FLFP with EVD-QNN classifier poses the ability to identify and categorize complicated patterns that leads to precise detections. Unlike previous IDS architectures, which utilize quantum-inspired neural networks or cryptography separately, this paper presents a highly integrated framework between the detection and security processes, with three main contributions: 1. EVD-augmented QNN feature embedding: In this paper, feature embedding using EVD is presented, allowing feature enrichment using QNN. 2. Heterogeneous Model Fusion using DSMMN-SLF: A novel score level fusion using double sigmoid median min-max normalization, termed DSMMN-SLF, is presented, allowing heterogeneous fusion between QNN and CNN. 3. Joint IDS + Cryptographic co-design: Unlike previous works, where these two processes were considered separately, this paper presents a highly integrated framework between IDS and EECC, allowing end-to-end secure inference and data transmission. 4. Lightweight Quantum-Inspired Model: Unlike previous quantum-inspired models, which were computationally intensive, this paper presents a highly accurate IDS framework with low computational complexity using hybridisation rather than quantum computing.

5. Proposes DSMMN-SLF for fusing the scores that captures numerous patterns and characteristics and produces significantly more accurate detected results. 6. Proposes EECC for ensuring the secure data transfer. 1. The developed EECC model attained equivalent security with less key size on comparing with conventional ECC model. The prevailing reviews on IDS works are outlined in Section II. The developed IDS for data security is detailed in section III. FLFP with EVD-QNN and LeNet are outlined in section IV. EECC-based encryption for data security is explained in section V. Sections VI and VII explained the results and conclusion.

II. RELATED WORK

A. RELATED WORKS

Lin et al. [18] used DL models in 2023 to forecast security-related indicators and derive appropriate security evaluation ratings. The security grade safety evaluation system was built using the ratings of these security assessments. Four distinct models, ResNet-50, VGG19, XceptionNet, and EfficientNet were used in the model training procedure. The EfficientNet model achieved a detection rate of 99.9% while using fewer computing resources in a single detection. Ultimately, the established security grade defense evaluation system gives a trustworthy and fast evaluation method for the network privacy. In 2024, Alrayes et al. [19] detailed how deploying and evaluating an IDS [32] built on DNNs may enhance network security. Given the critical importance of protecting computer networks from malicious activity, the study explored modern technical approaches for improving intrusion detection performance. The study also demonstrated how DNN-oriented IDS greatly boosts the network security by providing real-time threat monitoring and improved defense against emerging internet threats. The study lays the groundwork for future advancements in this area, highlighting the crucial role that DNN-based IDS play in modern cybersecurity practices. In 2024, Ibrahim et al. [20] investigated the application of ensemble learning to improve IDS efficacy in IoMT settings. Basic models such as KNN, MLP, XGB, RF, ADA, and LR were used for building the IDS in this work. A variety of indicators were used to assess the integrated ensemble model's performance. The outcomes of confusion matrix provided additional evidence of its effectiveness in precise categorization. In 2024, Rahman et al. [21] investigated the possibility of training ML models in NIDS using entirely synthetic data produced by GAN. This method greatly lessened the reliance on real-world data, opening the door to a more adaptable and morally sound model-building procedure. Further, 100% flawless scores were achieved on all measures for the BoT-IoT dataset. These results showed that this analysis's values work well, producing results that were on par with or better than those of earlier research. As a result, this research effectively replicated actual NID data, indicating possibilities for using generative data to cyber security. In 2023, Kadry et al. [22] introduced ECC and a QNN based on WCSO to precisely identify an attack and provide data security. The features in the network data that helped with accurate intrusion detection were selected using the WOA. The optimized QNN, which combined the FFBA with the WOA approach, was used to detect attacks. Sensitive data fetching required an encryption technique, for which, optimized ECC was used that could safely keep the data files with safety precautions. In the case that the data holder retains confidential information on the server, the file was encrypted utilizing the encryption mechanism. With a high accuracy of 98.5%, the QNN with WOA-oriented IDS system was a right option for actual time intrusion detection analysis. Thus, the study has shown that the proposed work offered better

secured data storage. A SecFedIDM-V1 utilizing BCT and BiLSTM-RNN was proposed by Mbaya et al. [23] in 2023. To create the suggested IDS model, the CIDDs was obtained, pre-processed, and divided for training, evaluation, and validation, respectively. Later, the created SecFedIDM-V1 was made available as a web application built on Python that records network packets and categorized attacks as either regular or specific. Bakro et al. [24] created an effective framework in 2023 that was applied for different network security solutions. Attacks were launched against each device connected to the network, and the overall attack rate was tracked. The cloud server was then protected from threats using a variety of technologies. The correctness of the results were tested using various principles at the end, and it was determined from each conclusion to what degree the resultants were superior to those of other works. In 2021, Nadeem et al. [25] presented an enhanced cloud IDS that incorporated the SMOTE to address the imbalanced data issue. A hybrid approach was suggested that combined 3 techniques: CS, IG, and PSO for feature selection. Lastly, a variety of attack types were detected and classified using the RF model. It was found that an IDS would function better if it had less informative characteristics. For many assessment measures, the simulation results performed noticeably better than other approaches suggested in the study.

B. REVIEW

The identification of more complex and covert attacks is found to be much complicated nowadays, as they imitate normal activity, which cannot be identified by existing detection systems. Rapid analysis and finding anomalies are computationally demanding due to the large amount and high complexity of contemporary data, particularly in remote and cloud systems. Furthermore, when illicit activity makes up a small portion of total traffic, the existing detection methods perform poorly due to dataset imbalances. It is crucial yet challenging to maintain high detection accuracy while maintaining lower false rates. Furthermore, as attackers are always changing their tactics, IDS must be scalable, dynamic, and able to learn from changing threats without much human involvement. The fact that these environments frequently have limited computing resources and differing security rules makes it challenging to design consistent and efficient IDS, which is another significant difficulty in attack detection for data security. Tracking private user information also raises concerns about security, thus methods that can identify breaches without jeopardizing confidentiality are required. Furthermore, attackers may influence ML-based detectors through methods like evasion attempts or data poisoning, underscoring the necessity of strong and resilient models. Moreover, the creation of efficient IDSs is made more difficult by the scarcity of labeled training datasets.

III. DEVELOPED INTRUSION DETECTION MODEL FOR DATA SECURITY

This work focuses on a novel deep learning (DL)-based intrusion detection system (IDS) integrated with data security

assurance by deploying an Enhanced Elliptical Curve Cryptography (EECC) model, as illustrated in Fig. 1.

Initially, the input data are preprocessed using a min–max normalization technique. Subsequently, a quantum-based hybrid detection model is introduced by combining the Fused Learning Particle Filter with Eigenvalue Decomposition–Quantum Neural Network (FLPF with EVD-QNN) and the LeNet classifier. The final intrusion detection decision is obtained using the DSMMN-based score-level fusion (DSMMN-SLF) method, which integrates the outputs of both classifiers into a single fused result.

Following intrusion detection, this work emphasizes secure data transmission. For this purpose, the proposed EECC scheme is employed to encrypt the detected data into a secure form prior to transmission.

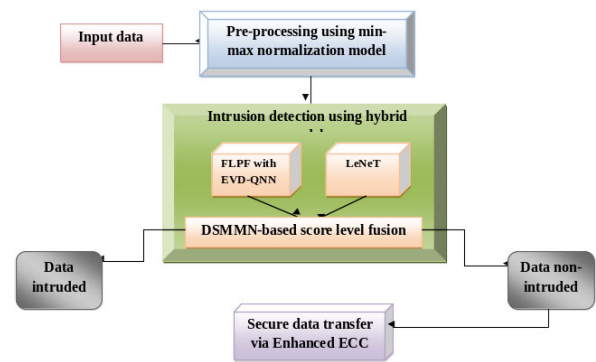


FIGURE 1. Framework of the proposed IDS model.

A. MIN-MAX NORMALIZATION FOR PREPROCESSING

Normalization techniques are essential to maintain data consistency and enable stable and accurate model training during preprocessing. In this work, the input dataset N is normalized using the min–max normalization approach.

In general, min–max normalization [26] is a widely adopted scaling technique that transforms features into a fixed range, typically between 0 and 1. The normalization formula is expressed as

$$N^n = \frac{N - \min(N)}{\max(N) - \min(N)}, \quad (1)$$

where $\min(N)$ and $\max(N)$ denote the minimum and maximum values of the dataset N , respectively, and N^n represents the normalized data.

The normalized samples N^n are then forwarded to the intrusion detection stage, which is accomplished using the proposed FLPF with EVD-QNN and LeNet hybrid model.

IV. FLPF WITH EVD-QNN AND LeNet FOR INTRUSION DETECTION

The normalized dataset N^n is fed into the FLPF with EVD-QNN and LeNet classifiers for detecting network intrusions, as depicted in Fig. 2.

The FLPF with EVD-QNN classifier is a modified variant of the conventional quantum neural network (QNN).

Since traditional QNN architectures suffer from structural limitations and reduced generalization capability, a novel QNN model with architectural refinements is proposed to improve detection accuracy and robustness.

Furthermore, the LeNet classifier is integrated with the FLPF with EVD-QNN model to further enhance detection performance by exploiting complementary spatial feature representations. Finally, the outputs of both classifiers are fused using the DSMMN-SLF technique to obtain a single optimized intrusion decision.

A. LeNET CLASSIFIER

LeNet [27] is a lightweight convolutional neural network (CNN) architecture originally developed for pattern recognition and detection tasks. Its convolutional layers are highly effective in extracting subtle spatial features while preserving hierarchical structures within the input data, thereby improving intrusion detection performance.

1) INPUT LAYER

The input layer receives the normalized samples N^n and forwards them to the first convolutional layer.

2) CONVOLUTIONAL LAYERS

The first convolutional layer employs the ReLU activation function and generates six feature maps, followed by average pooling for subsampling. The second convolutional layer also uses ReLU activation and produces sixteen feature maps, which are again subsampled using average pooling.

3) FLATTEN LAYER

The resulting feature maps are flattened into a one-dimensional vector to facilitate processing by the fully connected layers.

4) FULLY CONNECTED LAYERS

LeNet contains three fully connected layers. The first two layers employ the hyperbolic tangent (tanh) activation function with 120 and 84 neurons, respectively. The third layer uses the softmax activation function with 10 neurons for classification.

5) OUTPUT LAYER

The final output indicates whether the input sample corresponds to an intrusion or normal activity. The LeNet classifier output is denoted as o^{LNet} .

B. FLPF WITH EVD-QNN

Li et al. [28] suggested a generic QNN architecture that may be used for picture object categorization applications. A collection of quantum logic gates makes up this QNN. The QNN's quantum circuit is also frequently referred to as a parameterized quantum circuit (PQC) due to its changing parameters. Quantum circuits, like conventional circuits, may intuitively depict crucial information such as

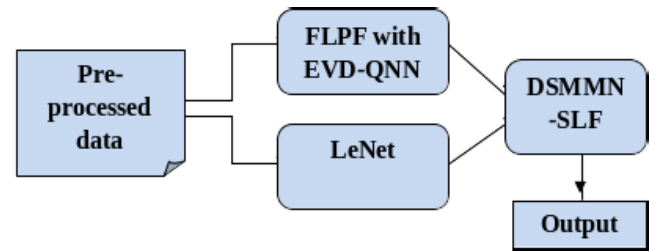


FIGURE 2. Proposed procedure for intrusion detection.

the complexity and structure of quantum algorithms. Qubits are the fundamental input unit in quantum circuits, and a number of quantum logic gates that represent unitary transformations are coupled to them in order to perform certain computing tasks. Quantum circuits are also referred to as quantum logic gate groups, and the fundamental building blocks for implementing them are quantum logic gates.

However, the existing quantum models include a restricted count of qubits and are vulnerable to high error values. This limits the size of QNN complexity. To overcome these issues, a high-quality feature map is proposed for intrusion detection.

Here, the FLPF with EVD-QNN classifier is proposed, where the output from the neural network is passed to a parameterized circuit. Moreover, the output from the flatten layer as well as the output from the EVD block are concatenated and the resultant is given as input to the fully connected (FC) layers.

Here, the FLPF with EVD-QNN classifier is trained with normalized data N^n for detecting intrusions.

The FLPF with EVD-QNN classifier contains six convolution layers, six batch normalization (BN) layers, two sigmoid operations, tanh, swish, Leaky ReLU, two max-pooling layers, average pooling, mixed pooling, scale dot product, and a flatten layer. Particularly, the EVD block is included, whose output is concatenated with the output of the flatten layer feature map.

The convolution layers in QNN obtain the input N^n by deploying filters or kernels to find patterns.

BN layers play an important role in training QNNs, especially quantized ones. By stabilizing the learning process, BN speeds up and improves the effectiveness of training.

1) SIGMOID AND TANH OPERATIONS

These activation functions make input values zero-centered by mapping them to a range around $[-1, 1]$, which aids learning and training.

2) POOLING FUNCTIONS

The spatial size of feature maps is decreased using max pooling, average pooling, and mixed pooling. This down-sampling technique increases computational efficiency and avoids overfitting by reducing the number of parameters.

3) SWISH AND LEAKY ReLU

These activation functions add non-linearity to QNNs, which aids the network's ability to recognize intricate patterns. The vanishing gradient issue in QNNs may be alleviated using Swish, a smoother activation function.

4) SCALE DOT PRODUCT

Scaling the dot product is essential for neural network performance and numerical stability, especially for large datasets.

5) FLATTEN LAYER

The flatten layer in QNN acts as a link between convolutional and FC layers. It converts the convolutional layers' multi-dimensional output into a one-dimensional vector so that dense or FC layers can handle it.

6) EIGENVALUE DECOMPOSITION (EVD)

The EVD is applied to the feature map derived from the flatten layer. This feature map is in matrix form. By applying EVD to this feature map, eigenvectors corresponding to their eigenvalues are obtained. These eigenvectors are merged with the flatten-layer feature map in QNN, which deploys quantum characteristics and improves feature representation.

7) EXAMPLE OF EVD TECHNIQUE

Let the flatten-layer feature map be in matrix form:

$$\mathbf{B} = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}. \quad (2)$$

For finding eigenvectors and eigenvalues, the following equation must be solved, where $\mathbf{B}$ denotes the matrix, χ the eigenvalue, and $\mathbf{V}$ the eigenvector:

$$\mathbf{B}\mathbf{V} = \chi\mathbf{V} \quad (3)$$

Let us begin by computing eigenvalues. For finding eigenvalues, Eq. (4) should be solved. In Eq. (4), $\mathbf{I}$ denotes the identity matrix.

$$|\mathbf{B} - \chi\mathbf{I}| = 0 \quad (4)$$

$$\left| \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix} - \chi \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right| = 0 \quad (5)$$

$$\left| \begin{bmatrix} 3-\chi & 2-0 \\ 1-0 & 4-\chi \end{bmatrix} \right| = 0 \quad (6)$$

$$\left| \begin{bmatrix} 3-\chi & 2 \\ 1 & 4-\chi \end{bmatrix} \right| = 0 \quad (7)$$

$$(3-\chi)(4-\chi) - 2 = 0 \quad (8)$$

$$12 - 3\chi - 4\chi + \chi^2 - 2 = 0 \quad (9)$$

$$\chi^2 - 7\chi + 10 = 0 \quad (10)$$

$$(\chi - 2)(\chi - 5) = 0 \quad (11)$$

$$\chi_1 = 2 \quad \text{and} \quad \chi_2 = 5 \quad (12)$$

Now let us find the related eigenvector for each eigenvalue.

For $\chi_1 = 2$, the eigenvector is found by solving Eq. (12):

$$(\mathbf{B} - \chi_1\mathbf{I})\mathbf{V}_1 = 0 \quad (13)$$

$$\begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix} - \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \quad (14)$$

$$\begin{bmatrix} 3-2 & 2 \\ 1 & 4-2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \quad (15)$$

$$\begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \quad (16)$$

Now reduce the matrix:

$$R_2 \leftarrow R_2 - R_1 = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \quad (17)$$

$$\begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (18)$$

$$x_1 + 2x_2 = 0 \quad (19)$$

$$x_1 = -2x_2 \quad (20)$$

The eigenvector corresponding to eigenvalue $\chi = 2$ is

$$\mathbf{V} = \begin{bmatrix} -2x_2 \\ x_2 \end{bmatrix} \quad (21)$$

Let $x_2 = 1$, then

$$\mathbf{V}_1 = \begin{bmatrix} -2 \\ 1 \end{bmatrix} \quad (22)$$

For $\chi = 5$,

$$(\mathbf{B} - \chi\mathbf{I})\mathbf{V}_2 = 0 \quad (23)$$

$$\begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \quad (24)$$

$$\begin{bmatrix} 3-5 & 2 \\ 1 & 4-5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \quad (25)$$

$$\begin{bmatrix} -2 & 2 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \quad (26)$$

Now reduce the matrix:

$$R_2 \leftarrow R_2 \div (-2) = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} \quad (27)$$

$$R_2 \leftarrow R_2 - R_1 = \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \quad (28)$$

$$\begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad (29)$$

$$x_1 - x_2 = 0 \quad (30)$$

$$x_1 = x_2 \quad (31)$$

The eigenvector corresponding to eigenvalue $\chi = 5$ is

$$\mathbf{V} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \Rightarrow \begin{bmatrix} x_2 \\ x_2 \end{bmatrix} \quad (32)$$

Let $x_2 = 1$, then

$$\mathbf{V}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad (33)$$

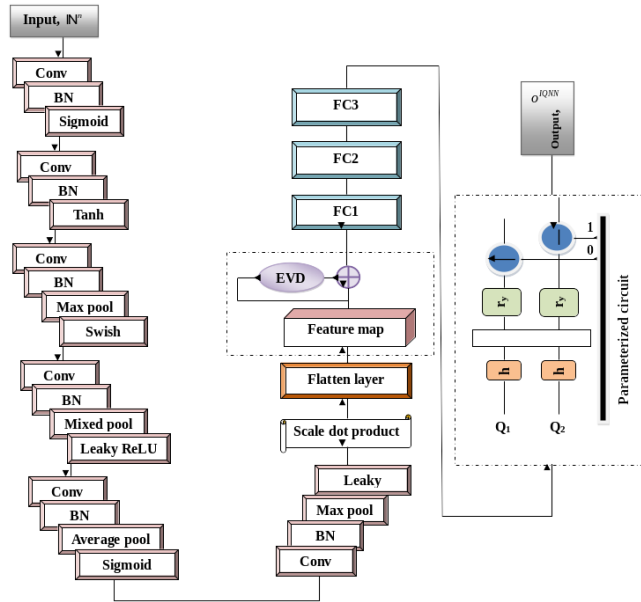


FIGURE 3. Proposed architecture for FLPF with EVD-QNN classifier.

Finally, the EVD of matrix $\mathbf{B}$ is computed as

$$\mathbf{B} = \mathbf{PDP}^{-1} = \begin{bmatrix} -2 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} -2 & 1 \\ 1 & 1 \end{bmatrix} \quad (34)$$

The proposed flatten-layer feature map is modeled as

$$\mathbf{B}' = \mathbf{B} \oplus \mathbf{PDP}^{-1} \quad (35)$$

The proposed FLPF with EVD-QNN classifier enables a high-quality feature map and captures complex patterns from the input data, leading to more precise intrusion detection. The classifier output is denoted by o^{QNN} . The architecture of the FLPF with EVD-QNN classifier is shown in Fig. 3.

Finally, the detection outcome is determined using the DSMMN-SLF method, which combines the outcomes of both FLPF with EVD-QNN and LeNet classifiers into a fused single output. However, it is important to clarify that the proposed QNN is a quantum-inspired neural network running on classical hardware, as opposed to quantum hardware. The parameterized quantum circuit is simulated classically using numerical simulations, where quantum gates are replaced by classical differentiable transformations. This is a common approach due to hardware limitations such as noise, decoherence, and scalability of current quantum hardware. Hence, it is important to clarify that the benefits seen in this study are due to quantum-inspired feature learning and non-linear transformation, rather than quantum parallelism. This design choice ensures computational feasibility while still leveraging the representational advantages of quantum-inspired transformations, thereby achieving a balance between performance and practical deployability.

C. DSMMN-SLF

The score-level fusion (SLF) method aggregates individual scores to obtain a unique score that is then used to make

a detection judgment. The existing SLF approach provides ratings on a variety of scales, which makes the fusion process difficult. Furthermore, it is unable to effectively capture the relationship between predictions.

Therefore, a new DSMMN-SLF method is proposed, which is computed based on min-max normalization as in Eq. (35), where o_i indicates the classifier score.

$$F_{\text{score}} = \sum_{i=1}^u o_i \quad (36)$$

$$o_i = \frac{o_i - \max(o_i)}{\max(o_i) - \min(o_i)} \quad (37)$$

According to DSMMN-SLF, the normalized score o_i^{new} is calculated as shown in Eq. (37), where q denotes a parameter, and σ and μ represent the standard deviation and mean, respectively.

$$o_i^{\text{new}} = \frac{1}{2} \left\{ \tanh \left(q \left(\frac{o_i - \mu(o_i)}{\sigma(o_i)} \right) \right) + 1 \right\} \quad (38)$$

As shown in Eq. (38), the DSMMN is evaluated based on o_i^{new} , where σ denotes the standard deviation.

$$(o_i^{\text{new}})' = \begin{cases} \frac{2(o_i^{\text{new}} - m_e)}{1 + \exp(-\omega(o_i^{\text{new}} - \delta))} - 1, & \text{if } o_i^{\text{new}} < \delta \\ \frac{o_i^{\text{new}} - \min(o_i^{\text{new}})}{\max(o_i^{\text{new}}) - \min(o_i^{\text{new}})}, & \text{if } o_i^{\text{new}} \geq \delta \end{cases} \quad (39)$$

$$\beta = K \sigma(o_i^{\text{new}}) \quad (40)$$

$$\delta = \text{Me}[\mu(o_i^{\text{new}}), o_i, \sigma(o_i^{\text{new}}), \sigma(o_i)] \quad (41)$$

Finally, the fusion score (fs) is formulated as in Eq. (41), which provides the detected output as intrusion or non-intrusion.

$$fs = w_1(o_1^{\text{new}})' + w_2(o_2^{\text{new}})' \quad (42)$$

$$w_i = \frac{\text{score}(o_i^{\text{new}})}{\sum \text{score}(o_i^{\text{new}})} \quad (43)$$

The numerous patterns and characteristics are captured by the DSMMN-SLF, which produces significantly more accurate detection results. Additionally, the implemented DSMMN-SLF provides suitable confidence scores.

The main benefit of the DSMMN-SLF scheme is due to its efficient approach in terms of normalizing and fusing heterogeneous classifier scores which may be at different levels. As mentioned previously, traditional score level fusion methods lack the ability to cope with scale problems and capturing non-linearities in relation to classifier scores. However, by means of the utilization of the double sigmoid function and the application of median normalization, DSMMN-SLF succeeds in achieving an increased separation of scores and retaining stability regardless of any data distribution.

Thus, the DSMMN-SLF score depicts the output as intrusion or non-intrusion. In the case of non-intrusion, the non-intruded data is encrypted using the EECC technique for secure data transfer.

V. EECC-BASED ENCRYPTION FOR DATA SECURITY

A. EECC METHOD

To enhance the IDS, creating a more robust security framework that secures the data through encryption standards is a critical aspect. Accordingly, this paper introduces EECC-based encryption for data security. A public-key cryptography technique called ECC [29] offers safe data transfer by performing key exchange, creating digital signatures, and encrypting and decrypting data. ECC is frequently used because of its effectiveness and its capacity to ensure security with reduced key sizes.

ECC is represented mathematically by Eq. (44), where p and X imply integers:

$$Z^2 = Y^3 + pY + X \quad (44)$$

The three keys used in ECC are:

- Secret key: v_k
- Public key: ρ_k
- Private key: η_k

The public key ρ_k is formulated as

$$\rho_k = \eta_k \times P_x \quad (45)$$

where P_x indicates a point on the curve.

The private key η_k is arbitrarily chosen from a set of j values. The secret key v_k is formulated as

$$v_k = \rho_k + \eta_k + P_x \quad (46)$$

Encryption: Eqs. (46) and (47) present the encryption formulation, where $G \in [0, m - 1]$, $E(\tau_1)$ and $E(\tau_2)$ denote ciphertexts, and O represents the original data.

$$E(\tau_1) = (G \times P_x) + v_k \quad (47)$$

$$E(\tau_2) = [O + (G \times \rho_k)] + v_k \quad (48)$$

Decryption: The decryption formulation is given by

$$O = E(\tau_2) - \eta_k (E(\tau_1) - v_k) - v_k \quad (49)$$

Although conventional ECC provides strong encryption, it introduces a correction factor that must be considered during decryption. Therefore, a novel EECC model is proposed for secure data transfer.

1) PROPOSED EECC

According to EECC, the secret key v_k is computed as

$$v_k = \frac{[(\rho_k \eta_k) + P_x] + [(\rho_k P_x) + \eta_k]}{2[(\eta_k P_x) + \rho_k]} \quad (50)$$

Proposed Encryption: Encryption is performed using ρ_k , v_k , and η_k as

$$E(\tau_1) = \frac{(\kappa P_x) + (\kappa \rho_k) + (\kappa \eta_k)}{P_x} + \frac{\eta_k}{2} \quad (51)$$

$$E(\tau_2) = O + [(\rho_k + \eta_k + P_x) - v_k] + E(\tau_1) \quad (52)$$

where κ is a random integer in $[0, l - 1]$.

Proposed Decryption:

$$O = E(\tau_2) - [(\rho_k + \eta_k + P_x) - v_k] - E(\tau_1) \quad (53)$$

The developed EECC model attains equivalent security with a smaller key size compared to conventional ECC. Furthermore, the proposed EECC reduces computational overhead and improves performance accuracy. In addition, it ensures confidentiality and minimizes the risk of threats during data transmission.

B. SECURITY ANALYSIS OF EECC

The security of the proposed Enhanced Elliptic Curve Cryptography (EECC) scheme is analysed with respect to common cryptographic attack models as follows:

- **Brute Force Attack (BFA):** The complexity of the combined key formulation, as defined in Eq. (50), significantly enlarges the effective key space. This increased key-space complexity makes exhaustive search attacks computationally infeasible, thereby enhancing resistance against brute force attacks.
- **Chosen Ciphertext Attack (CCA) and Known Plaintext Attack (KPA):** The addition of a random parameter κ to the encryption function introduces non-determinism to the generation of ciphertext. This means that two identical plaintext inputs to the encryption function can result in two different ciphertexts, making it difficult to derive statistical correlations between them, which an attacker might use to deduce information through ciphertext or known plaintext attacks.
- **Man-in-the-Middle Attack (MIMA):** The interdependent relationship between the public, private, and secret keys in EECC creates a tightly coupled relationship. This relationship prevents an attacker from reconstructing the keys without the full knowledge of the underlying key generation mechanism. Compared with the conventional ECC, the proposed non-linear coupling of the keys increases the resistance against interception-based attacks.

Despite the fact that the EECC proposal exhibits robust empirical resilience to popular attacks like BFA, CCA, CPA, and MIMA, it should be recognized that a rigorous cryptographic proof of security using standard hardness assumptions like ECDLP has not been fully developed in this study. In this analysis, the emphasis has been put on empirical testing and performance comparisons. A rigorous theoretical proof of security will be taken into consideration as an avenue for future research.

VI. RESULTS AND DISCUSSION

A. SIMULATION PROCEDURE

The implemented FLFP with EVD-QNN+LNt model for IDS was accomplished using Python. In addition, the simulation was conducted on a system equipped with a 64-bit operating system, an x64-based processor, and 16.0 GB (15.7 GB usable) of RAM. The FLFP with EVD-QNN+LNt model was evaluated against QNN+ECC [22], QAE [30], PolyNet, LinkNet, Bi-LSTM, SqueezeNet, and LeNet models.

The UNSW-NB15 dataset was used for the analysis [31]. For this purpose, the dataset is divided into training, validation, and testing sets using a stratified sampling method, where 70% of the data is used for training, 15% for validation, and 15% for testing. All experiments were carried out 10 times, and the results presented here are averages of all performance metrics. To ensure reproducibility, random seeds were set for NumPy, TensorFlow/PyTorch, and system initialization. Hyperparameters such as learning rate, batch size, and optimizer were kept constant throughout all experiments.

1) DATASET DESCRIPTION

The raw network packets of the UNSW-NB15 dataset were generated using the IXIA PerfectStorm tool in the Cyber Range Lab of UNSW Canberra to create a hybrid of real modern normal activities and synthetic contemporary attack behaviours. The *tcpdump* tool was utilised to capture 100 GB of raw traffic (PCAP files). The dataset contains nine attack categories, namely Fuzzers, Analysis, Backdoors, DoS, Exploits, Generic, Reconnaissance, Shellcode, and Worms. The Argus and Bro-IDS tools were employed, and twelve algorithms were developed to generate a total of 49 features along with the class labels. The amount of data in the set was limited to 9,000 with balanced classes, as this was done with the aim of controlled experimental evaluation and convergence during the training process. Although the real-world IDS data sets are naturally unbalanced, balancing the classes allows for unbiased comparison across models, preventing bias towards the majority class. To evaluate the models' generality, further experiments with unbalanced classes are proposed for future studies. However, the chosen data set maintains the diversity of attack classes as in the UNSW-NB15 data set. It is important to note that the data set was balanced, meaning that the number of data in each class was equal, with 9,000 data points in total.

For the proposed analysis, 9000 samples were used with two classes: 0 (non-attack) and 1 (attack). Each class consisted of 4500 samples. The learning rate was set to 0.001 for both QNN and LeNet models.

B. QUANTITATIVE PERFORMANCE EVALUATION

The performance of the proposed IDS using FLPF with EVD-QNN+LNt is illustrated in Fig. 4, Fig. 5, and Fig. 6 across multiple evaluation metrics. The proposed approach is compared with QNN+ECC, QAE, PolyNet, LinkNet, Bi-LSTM, SqueezeNet, and LeNet models. For a fair comparison, it is important to note that all baseline models were re-implemented and fine-tuned with optimal hyperparameters as presented in the literature. Therefore, it is evident that the improvement in the proposed model's performance is due to architectural benefits and not suboptimal configuration of baseline models.

In Fig. 4(c), the precision achieved by FLPF with EVD-QNN+LNt is 0.97 at 90% TD, which is higher than all

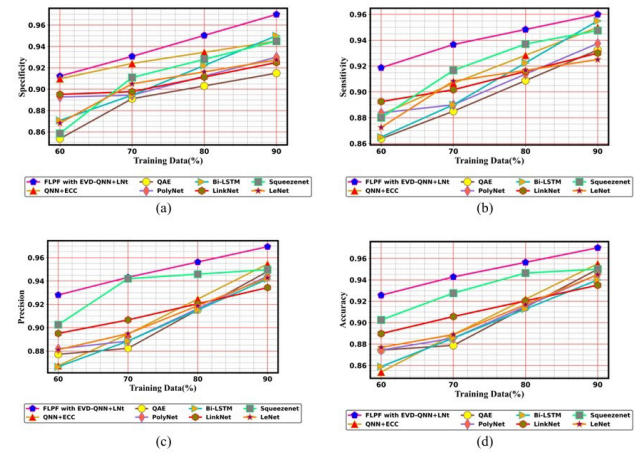


FIGURE 4. Performance of FLPF with EVD-QNN+LNt over prevailing models for IDS on (a) Specificity (b) Sensitivity (c) Precision (d) Accuracy.

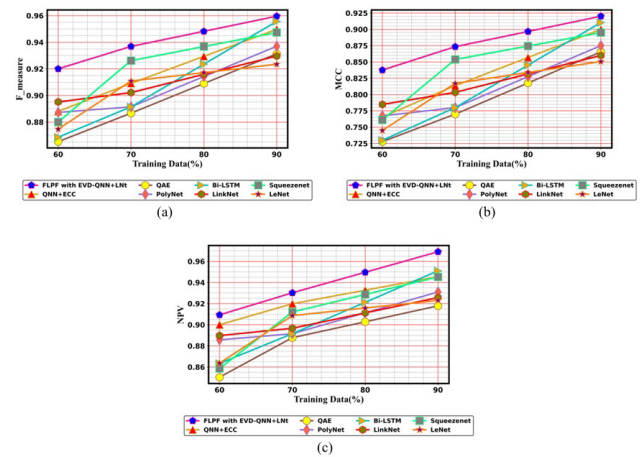


FIGURE 5. Performance of FLPF with EVD-QNN+LNt over prevailing models for IDS on (a) F measure (b) MCC (c) NPV.

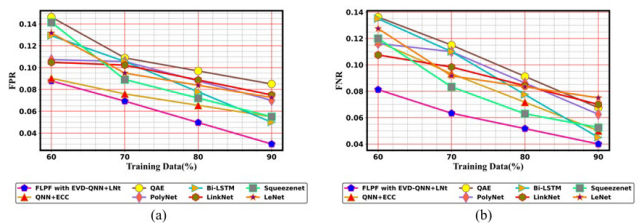


FIGURE 6. Performance of FLPF with EVD-QNN+LNt over prevailing models for IDS on (a) FPR (b) FNR.

the existing models. The proposed FLPF with EVD-QNN classifier generates high-quality feature maps that capture complex and fine-grained patterns from the input data, resulting in improved intrusion detection capability. This also leads to a low false positive rate (FPR) of 0.03 compared to the other models. Furthermore, the neutral scores obtained using FLPF with EVD-QNN+LNt outperform the existing techniques, demonstrating the effectiveness of the proposed IDS.

To avoid any bias or reproducibility in the assessment phase, the baseline models have been designed and optimized through the use of identical experimental parameters. This includes the setting of proper learning rates, batch sizes, and optimization parameters. All these were chosen from previous studies that used optimal techniques. The learning rates considered were between 0.0001 and 0.001, whereas Adam was used as an optimizer for all the models. Moreover, all the models were trained for appropriate number of epochs and early stopping methods were applied to prevent overfitting. By optimizing the models in the identical manner, any improvement in the proposed model is solely due to its effective architecture.

C. ABLATION ANALYSIS

The ablation study using suggested FLPP with EVD-QNN+LNt for IDS is provided in Table 1 for TD = 80%. The ablation of developed FLPP with EVD-QNN+LNt is done over LNt with existing QNN, FLPP with EVD-QNN+LNt with existing SLF, FLPP with EVD-QNN+LNt with no SLF and FLPP with EVD-QNN+LNt with no normalization. The presented FLPP with EVD-QNN+LNt shows better accuracy of 0.95638, while, LNt with existing QNN, FLPP with EVD-QNN+LNt with existing SLF, FLPP with EVD-QNN+LNt with no SLF and FLPP with EVD-QNN+LNt with no normalization gained less accuracy scores of 0.931662, 0.915825, 0.902357 and 0.90404. The proposed FLPP with EVD-QNN classifier enables feature map with high quality and leads to fine depiction of complicated patterns and associations from the provided data. The numerous patterns and characteristics are captured by the DSMMN-SLF, which produces significantly more accurate detected results. Further, the proposed EECC lessens the computing overhead and enhanced the accuracy of performance. This enhanced the NPV of 0.949652 using FLPP with EVD-QNN+LNt, while, LNt with existing QNN, FLPP with EVD-QNN+LNt with existing SLF, FLPP with EVD-QNN+LNt with no SLF and FLPP with EVD-QNN+LNt with no normalization gained less NPV values. Thus, suggested FLPP with EVD-QNN+LNt is well appropriate for identifying the intrusions.

D. ABLATION ANALYSIS INTERPRETATION

The observed performance improvement of the proposed model can be attributed to the synergistic contribution of its key components, as detailed below:

- **EVD Module:** The Eigenvalue Decomposition (EVD) module enhances feature representation by capturing the dominant spectral components of the feature space. This enables improved separability between normal and attack patterns, thereby increasing the discriminative capability of the model.
- **DSMMN-SLF:** The proposed Double Sigmoid Median Min-Max Normalization-based Score Level Fusion (DSMMN-SLF) improves robustness by effectively normalizing classifier outputs. This mitigates scale

inconsistencies arising from heterogeneous classifiers and ensures stable and reliable fusion decisions.

- **Hybrid Fusion (QNN + CNN):** The integration of quantum-inspired neural networks (QNN) and convolutional neural networks (CNN) enables complementary feature extraction. Specifically, QNN captures global and complex feature interactions, while CNN effectively extracts local spatial patterns, leading to enhanced overall detection performance.

Therefore, the removal of any of these components results in degraded performance, as evidenced in the ablation study, highlighting the importance of each module in the proposed architecture.

E. STATISTICAL ANALYSIS

The statistical analysis using FLPP with EVD-QNN+LNt is specified in Table 2 that reveals the accuracy of proposed work. The analysis shows the efficacy of FLPP with EVD-QNN+LNt over QNN+ECC, QAE, PolyNet, LinkNet, Bi-LSTM, Squeezenet and LeNet. In case of mean, a better accuracy of 0.948696 is obtained due to the improvements in proposed FLPP with EVD-QNN+LNt. While, QNN+ECC, QAE, PolyNet, LinkNet, Bi-LSTM, Squeezenet and LeNet attained slightest accuracies of 0.90492, 0.904385, 0.905035, 0.912707, 0.899314, 0.935426 and 0.906939 correspondingly. The implemented FLPP with EVD-QNN+LNt classifier thus provided enhanced detection accuracy.

F. ATTACK ANALYSIS

The analysis showing the impact of attacks while applying EECC-based encryption over existing encrypting approaches is given in Fig. 7. The considered attacks include BFA, CCA, CPA, KCA and MIMA. Here, BFA and MIMA are examined based on key breaking time, while, CCA, CPA and KCA are examined based on correlation. For ensuring data security, the key breaking time needs to be high, while, correlation between original and encrypted data must be less. On analyzing BFA attack, the key breaking time is higher for data variation of 100% using proposed EECC. For 100% data variation, the EECC-based encryption attained higher key breaking time of 39 sec, while, AES, Elgamal, blowfish, ECC and RSA attained less key breaking time. For CPA attack, the proposed EECC-based encryption displays less correlation value of 0.135 at data variation of 100%, while, AES, Elgamal, blowfish, ECC and RSA attained high correlation values. This confirms that the correlation between original and encrypted data are less, thus ensuring better data security. Likewise, for all attacks, the correlation is less while applying EECC-based encryption over other encrypting methods.

G. ANALYSIS ON TIME COMPLEXITY

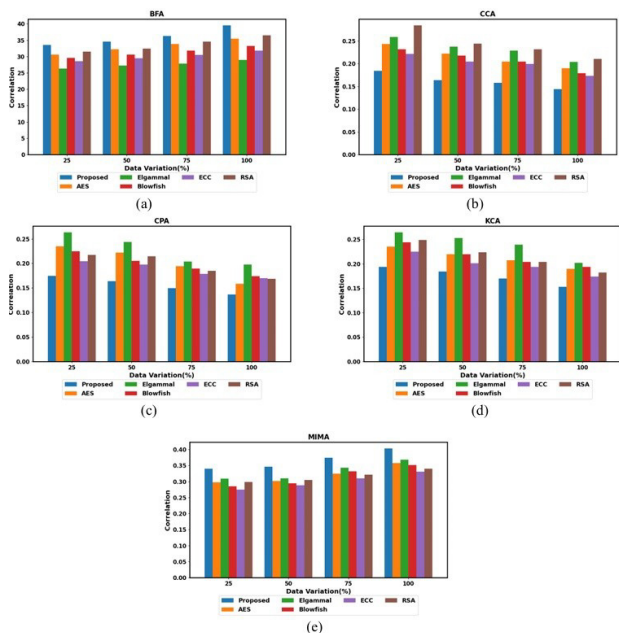
Fig. 8 expresses the time utilized for encryption and decryption while securing the non-intruded data. The time

TABLE 1. Ablation study using FLPF with EVD-QNN+Lnt model for IDS.

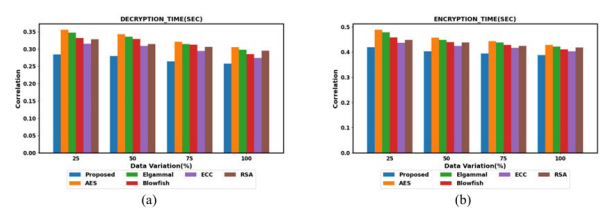
Metrics	FLPF with EVD-QNN+Lnt	Lnt with existing QNN	FLPF with EVD-QNN+Lnt with existing SLF	FLPF with EVD-QNN+Lnt with no SLF	FLPF with EVD-QNN+Lnt with no normalization
NPV	0.949652	0.922264	0.917565	0.920321	0.908494
Specificity	0.950347	0.922071	0.919142	0.923267	0.910479
Accuracy	0.956380	0.931662	0.915825	0.902357	0.904040
FNR	0.049653	0.077929	0.080858	0.076733	0.089521
F-measure	0.948237	0.926968	0.918517	0.914688	0.908422
MCC	0.896800	0.854166	0.835486	0.826309	0.814841
Sensitivity	0.948333	0.926875	0.917500	0.912917	0.907292
FPR	0.043620	0.068338	0.084175	0.097643	0.095960
Precision	0.956266	0.932335	0.918442	0.906673	0.906670

TABLE 2. Statistical comparison of FLPF with EVD-QNN+Lnt and existing deep learning models.

Metrics	FLPF with EVD-QNN+Lnt	QNN+ECC	QAE	PolyNet	LinkNet	Bi-LSTM	SqueezeNet	LeNet
Max	0.970000	0.955000	0.950000	0.945000	0.935000	0.940000	0.950000	0.945000
Min	0.925641	0.853846	0.874359	0.874359	0.889744	0.858974	0.902564	0.876923
SD	0.016429	0.037628	0.030568	0.027493	0.016820	0.030223	0.019145	0.026342
Median	0.949571	0.905417	0.896591	0.900391	0.913043	0.899141	0.944571	0.902917
Mean	0.948696	0.904920	0.904385	0.905035	0.912707	0.899314	0.935426	0.906939

**FIGURE 7.** Analysis using EECC over prevailing encrypting models for attacks namely, (a) BFA (b) CCA (c) CPA (d) KCA (e) MIMA.

(in sec) utilized by EECC is lesser than AES, Elgamal, blowfish, ECC and RSA methods. The time utilized for encrypting the transferred data is 0.39 sec, which is less than the time utilized for encryption by existing AES, Elgamal, blowfish, ECC and RSA methods. The decrypting time is

**Fig.8.** Analysis using EECC over prevailing encrypting models for (a) Decryption time and (b) Encryption time**FIGURE 8.** Analysis using EECC over prevailing encrypting models for (a) Decryption time and (b) Encryption time.

about 0.26 sec, which is less than the time utilized for decryption by existing AES, Elgamal, blowfish, ECC and RSA methods. Thus, the developed EECC model attained equivalent security with less key size on comparing with conventional models. This in turn lessened the computing overhead and lessened the time utilization for encryption and decryption.

H. ANALYSIS ON KEY SENSITIVITY AND LATENCY

Fig. 9 shows the analysis on key sensitivity and latency. When distinguished to prevailing AES, Elgamal, blowfish, ECC and RSA methods, the proposed EECC has achieved less correlation values for key sensitivity. Particularly, at data variation of 100%, the proposed EECC attained less correlation of 0.16, which ensures that the correlation between original and encrypted data are less. The latency given in Fig. 9(b), is computed by adding encrypting time as well as decrypting time. Thus, latency should be less

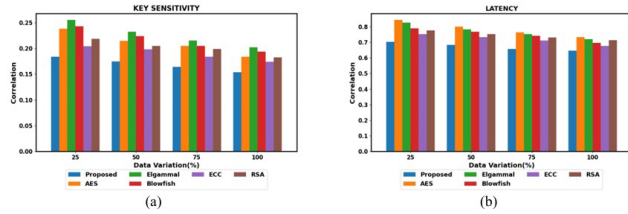


FIGURE 9. Analysis using EECC over prevailing encrypting models for (a) Key sensitivity and (b) Latency.

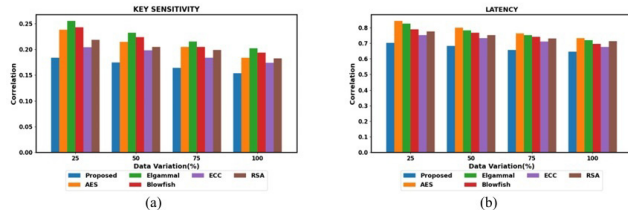


FIGURE 10. ROC analysis using FLPF with EVD-QNN+Lnt over prevailing models for IDS.

to ensure less time consumption. Regarding latency, the proposed EECC obtained less value of 0.645, while, AES, Elgamal, blowfish, ECC and RSA methods achieved high latency values.

I. ROC ANALYSIS

The ROC study for IDS and data security is given in Fig. 10. This study makes examination on FPR vs TPR. When FPR is 0.005, the TPR for QNN+ECC, QAE, PolyNet, LinkNet, Bi-LSTM, SqueezeNet and LeNet are minimal than the TPR achieved by proposed FLPF with EVD-QNN+Lnt classifier. For all FPRs, the proposed FLPF with EVD-QNN+Lnt shows high TPR values. Thus, the proposed EECC lessens the computing overhead and enhanced the accuracy of safer data transfer.

J. PARAMETER SENSITIVITY ANALYSIS

The performance of the proposed DSMMN-SLF mechanism is influenced by key parameters, namely q , ω , and δ . A sensitivity analysis is conducted to evaluate the impact of these parameters on the overall model performance.

The analysis reveals the following observations:

- **Effect of q :** The parameter q controls the degree of non-linearity in the transformation function. Moderate values of q introduce sufficient non-linearity to enhance discriminative capability, while excessively high values may lead to instability and over-amplification of score variations.
- **Effect of δ :** The parameter δ governs the thresholding behaviour in the fusion process. It determines the transition point for score normalization, thereby influencing classification sensitivity and decision boundaries.
- **Effect of ω :** The parameter ω regulates the steepness of the sigmoid function. Higher values increase the

sharpness of the transition, improving separability but potentially reducing smoothness, whereas lower values ensure gradual transitions and improved stability.

The experimental results indicate that the proposed DSMMN-SLF framework maintains stable performance across a wide range of parameter values, demonstrating its robustness and reliability for intrusion detection tasks.

K. NOTATION AND SYMBOL DEFINITION

For clarity, the key symbols used in the DSMMN-SLF fusion process are summarised as follows:

- o_i : Output score of the i^{th} classifier
- μ : Mean of the classifier scores
- σ : Standard deviation of the classifier scores
- q, ω, δ : Parameters controlling the non-linear transformation and fusion process
- f_s : Final fused score used for intrusion classification

VII. CONCLUSION

This work focused on a new DL-based IDS model along with data security assurance by deploying enhanced ECC model. Initially, the input data was pre-processed using min-max normalizing technique. Subsequently, Quantum based Hybrid model was introduced that combined the models like FLPF with EVD-QNN and LeNet models. Here, the detection results were determined by DSMMN-SLF. Subsequently to the intrusion detection, this work concentrated on secured data transmission, for which EECC was proposed that encrypted the data to a secured form. On analyzing, the precision of detecting the intrusions using FLPF with EVD-QNN+Lnt was 0.97, which is high when comparing to existing QNN+ECC, QAE, PolyNet, LinkNet, Bi-LSTM, SqueezeNet and LeNet that attained lesser precision scores. Moreover, the time utilized for encrypting the transferred data was 0.39 sec, which is less than the time utilized for encryption by existing AES, Elgamal, blowfish, ECC and RSA methods. In future, features can be extracted and then classified for identifying the intrusions.

LIMITATIONS AND FUTURE WORK

Despite the excellent detection performance and security integration, there are a few limitations of the current study that must be mentioned. First, it must be mentioned that the experimental evaluation of the current study is performed on a balanced subset of the UNSW-NB15 dataset. Although it is an effective approach for benchmarking, it is also important to mention that the current study might not effectively tackle the intrinsic class imbalance of network traffic, as it is a rare occurrence in network intrusion scenarios. Second, it must be mentioned that the current study uses a quantum neural network (QNN) component, which is a quantum-inspired approach, meaning it is a classical implementation of a QNN. This is a limitation of the current study, as it might not effectively use the power of quantum computing, which might be more effective for QNNs.

Third, while the proposed EECC method is shown to have empirical robustness against various attack models such as BFA, CPA, CCA, and MIMA, the security of EECC is still largely heuristic. A formal cryptographic proof, especially under general security assumptions such as the Elliptic Curve Discrete Logarithm Problem (ECDLP), is not fully supported in this paper. Thus, the theoretical security of EECC needs further investigation. Furthermore, while various components such as EVD-based feature enhancement, DSMMN-SLF fusion, and hybrid QNN-CNN architecture were incorporated, these could have implications on the complexity of the method, which could affect its computational efficiency, especially in resource-constrained environments. Due to computational constraints, a full quantitative evaluation on imbalanced data is left for future work. However, based on the robustness of the DSMMN-SLF fusion mechanism and the stability observed in multi-run experiments, the proposed model is expected to maintain performance under skewed distributions.

The future direction of this work involves:

- (i) testing this model on large-scale, real-world, and highly imbalanced datasets,
- (ii) exploring this model on new quantum hardware platforms to demonstrate quantum advantage,
- (iii) constructing a formal security proof of EECC under well-established cryptographic models, and
- (iv) optimising this architecture to achieve low-latency and efficient deployment in real-world networks.

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REFERENCES

- [1] E.-U.-H. Qazi, T. Zia, M. H. Faheem, K. Shahzad, M. Imran, and Z. Ahmed, "Zero-touch network security (ZTNS): A network intrusion detection system based on deep learning," *IEEE Access*, vol. 12, pp. 141625–141638, 2024.
- [2] C. Park, J. Lee, Y. Kim, J.-G. Park, H. Kim, and D. Hong, "An enhanced AI-based network intrusion detection system using generative adversarial networks," *IEEE Internet Things J.*, vol. 10, no. 3, pp. 2330–2345, Feb. 2023.
- [3] O. K. ChandraUmakantham, S. Gajendran, and S. Marappan, "Enhancing intrusion detection through federated learning with enhanced ghost_BiNet and homomorphic encryption," *IEEE Access*, vol. 12, pp. 24879–24893, 2024.
- [4] T. Kim and W. Pak, "Early detection of network intrusions using a GAN-based one-class classifier," *IEEE Access*, vol. 10, pp. 119357–119367, 2022.
- [5] K. He, W. Zhang, X. Zong, and L. Lian, "Network intrusion detection based on feature image and deformable vision transformer classification," *IEEE Access*, vol. 12, pp. 44335–44350, 2024.
- [6] L. Almuqren, H. Alqahtani, S. S. Aljameel, A. S. Salama, I. Yaseen, and A. A. Alneil, "Hybrid metaheuristics with machine learning based botnet detection in cloud assisted IoT," *IEEE Access*, vol. 11, pp. 115668–115676, 2023.
- [7] X. Huang, J. Liu, Y. Lai, B. Mao, and H. Lyu, "EEFED: Personalized federated learning of execution & evaluation dual network for CPS intrusion detection," *IEEE Trans. Inf. Forensics Security*, vol. 18, pp. 41–56, 2023.
- [8] A. Algarni, Z. Ahmad, and M. A. Ala'anzy, "An edge computing-based and threat behavior-aware smart prioritization framework for cybersecurity intrusion detection and prevention of IEDS in smart grids with integration of modified lgbm and one class-SVM models," *IEEE Access*, vol. 12, pp. 104948–104963, 2024.
- [9] Z. Á. Mann, F. Kunz, J. Laufer, J. Bellendorf, A. Metzger, and K. Pohl, "RADAR: Data protection in cloud-based computer systems at run time," *IEEE Access*, vol. 9, pp. 70816–70842, 2021.
- [10] A. B. Z. Mahmoodi, S. Sheikhi, E. Peltonen, and P. Kostakos, "Autonomous federated learning for distributed intrusion detection systems," *IEEE Access*, vol. 11, pp. 121325–121339, 2023.
- [11] D. Javeed, T. Gao, M. S. Saeed, and P. Kumar, "Intrusion detection for edge-envisioned smart agriculture," *IEEE Internet Things J.*, vol. 11, no. 16, pp. 26866–26876, 2023.
- [12] A. Yahyaoui, T. Abdellatif, S. Yangui, and R. Attia, "READ-IoT: Reliable event and anomaly detection framework," *IEEE Access*, vol. 9, pp. 24168–24186, 2021.
- [13] A. H. Janabi, T. Kanakis, and M. Johnson, "Overhead reduction technique for SDN-based IDS," *IEEE Access*, vol. 10, pp. 66481–66491, 2022.
- [14] S. Lin, C. Feng, T. Jiang, and H. Jing, "Evaluation of network security grade protection combined with deep learning for intrusion detection," *IEEE Access*, vol. 11, pp. 130990–131000, 2023.
- [15] O. Arreche, T. R. Guntur, J. W. Roberts, and M. Abdallah, "E-XAI: Evaluating event and anomaly explainable AI for IDS," *IEEE Access*, vol. 12, pp. 23954–23988, 2024.
- [16] R. Ishibashi, K. Miyamoto, C. Han, T. Ban, T. Takahashi, and J. Takeuchi, "Generating labeled datasets for unified IDS," *IEEE Access*, vol. 10, pp. 53972–53986, 2022.
- [17] N. Tran, H. Chen, J. Bhuyan, and J. Ding, "Data curation and quality evaluation for ML-based cyber IDS," *IEEE Access*, vol. 10, pp. 121900–121923, 2022.
- [18] S. Gaba et al., "A systematic analysis of enhancing cyber security using deep learning for cyber physical systems," *IEEE Access*, vol. 12, pp. 6017–6035, 2024, doi: [10.1109/ACCESS.2023.3349022](https://doi.org/10.1109/ACCESS.2023.3349022).
- [19] F. Alrayes, M. Zakariah, S. Amin, Z. Khan, and J. Alqurni, "Network security enhanced with deep neural network IDS," *Comput., Mater. Continua*, vol. 80, no. 1, pp. 1457–1490, 2024.
- [20] M. M. Alani, A. I. Awad and E. Barka, "A hybrid ensemble learning-based intrusion detection system for the Internet of Things," in *Proc. IEEE Int. Conf. Cyber Secur. Resilience (CSR)*, London, U.K., 2024, pp. 1–8, doi: [10.1109/CSR61664.2024.10679427](https://doi.org/10.1109/CSR61664.2024.10679427).
- [21] S. Rahman, S. Pal, S. Mittal, T. Chawla, and C. Karmakar, "SYN-GAN: Robust intrusion detection using GAN-based synthetic data," *Internet Things*, vol. 26, p. 101212, Jul. 2024.
- [22] H. Kadry, A. Farouk, E. A. Zanaty, and O. Reyad, "Intrusion detection using optimized QNN and ECC," *Alexandria Eng. J.*, vol. 71, pp. 491–500, May 2023.
- [23] E. B. Mbaya, E. Adetiba, J. A. Badejo, J. S. Wejin, O. Oshin, and O. Isife, "SecFedIDM-V1: Secure federated IDS with blockchain," *IEEE Access*, vol. 11, pp. 116011–116025, 2023.
- [24] M. Bakro, R. R. Kumar, A. A. Alabrah, Z. Ashraf, M. N. Ahmed, M. Shameem, and A. Abdelsalam, "An improved design for a cloud intrusion detection system using hybrid features selection approach with ML classifier," *IEEE Access*, vol. 11, pp. 64228–64247, 2023.
- [25] M. Nadeem, A. Arshad, S. Riaz, S. S. Band, and A. Mosavi, "Intercepting cloud network attacks via IDS," *IEEE Access*, vol. 9, pp. 152300–152309, 2021.
- [26] J. Prasetyo, N. A. Setiawan, and T. B. Adji, "Improving normalization method of higher-order neural networks," in *Proc. E3S Web Conf.*, vol. 200. EDP Sciences, 2020, Paper 02016.
- [27] S. Balasubramaniam, Y. Velmurugan, D. Jaganathan, and S. Dhanasekaran, "Modified LeNet CNN for breast cancer diagnosis," *Diagnostics*, vol. 13, no. 17, p. 2746, 2023.
- [28] Y. Li, R. Liu, X. Hao, R. Shang, P. Zhao, and L. Jiao, "EQNAS: Evolutionary quantum neural architecture search," *Neural Netw.*, vol. 168, pp. 471–483, Nov. 2023.
- [29] D. V. K. Vengala, D. Kavitha, and A. P. S. Kumar, "Three-factor authentication with modified ECC," *Complex Intell. Syst.*, vol. 9, no. 3, pp. 2915–2928, 2023.
- [30] M. Hdaib, S. Rajasegarar, and L. Pan, "Quantum deep learning-based anomaly detection," *Quantum Mach. Intell.*, vol. 6, no. 1, p. 26, 2024.
- [31] UNSW Canberra Cyber Range Lab. (2015). *UNSW-NB15 Dataset*. [Online]. Available: <https://research.unsw.edu.au/projects/unsw-nb15-dataset>
- [32] O. A. Alzubi, J. A. Alzubi, T. M. Alzubi, and A. Singh, "Quantum mayfly optimization with encoder-decoder driven LSTM networks for malware detection and classification model," *Mobile Netw. Appl.*, vol. 28, no. 2, pp. 795–807, Apr. 2023.



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